Kleisli Composition

Season: 7 · Advanced FP · Category: Functional · Level: Advanced

Intent

- Compose A → F<B> arrows.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Effectful function pipelines.

When to Use

- Serious FP architecture in Java.
- The design force named in Intent shows up more than once.

When Not to Use

- CRUD apps that don't need effect algebras.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.functional.kleisli;

import com.buildwithwaqar.patterns.functional.typeclass.Kind;
import com.buildwithwaqar.patterns.functional.typeclass.Monad;

import java.util.function.Function;

/** Kleisli arrow A =&gt; F&lt;B&gt; — composable effectful functions. */
public record Kleisli<F, A, B>(Function<A, Kind<F, B>> run) {
    public Kind<F, B> apply(A a) { return run.apply(a); }

    public <C> Kleisli<F, A, C> andThen(Monad<F> M, Kleisli<F, B, C> next) {
        return new Kleisli<>(a -> M.flatMap(run.apply(a), next.run));
    }

    public static <F, A, B> Kleisli<F, A, B> of(Function<A, Kind<F, B>> f) {
        return new Kleisli<>(f);
    }
}
```

Modern Java 25 / Functional

- Witness types emulate higher-kinded types.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: functional/kleisli/
```

Pitfalls

- Encodings are educational — production may use libraries.
- Tagless Final needs Kind encoding without HKTs.

Interview Q&A

- Q: When to pick Kleisli Composition? A: When you need testable effect boundaries or pure state threading.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Compose A → F<B> arrows.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Effectful function pipelines. [0:40] Intent — State the design force in one sentence: Compose A → F&...